

PROJECT BUTTON HR-V0 / HR-30 ELECTRICAL V2.1

Connected design intent generated from one model; unresolved hardware and pin maps remain explicit release blockers.

01 01 ac pe power

File: 01_ac_pe_power.kicad_sch

02 02 hrv0 safety

File: 02_hrv0_safety.kicad_sch

03 03 hrv0 distribution

File: 03_hrv0_distribution.kicad_sch

04 04 control watchdog

File: 04_control_watchdog.kicad_sch

05 05 hr30 battery precharge

File: 05_hr30_battery_precharge.kicad_sch

06 06 hr30 safety distribution

File: 06_hr30_safety_distribution.kicad_sch

07 07 left leg rs485

File: 07_left_leg_rs485.kicad_sch

08 08 right leg rs485

File: 08_right_leg_rs485.kicad_sch

09 09 upper bus a

File: 09_upper_bus_a.kicad_sch

10 10 upper bus b

File: 10_upper_bus_b.kicad_sch

11 11 sensors sync

File: 11_sensors_sync.kicad_sch

12 12 sensor adc imu

File: 12_sensor_adc_imu.kicad_sch

13 13 sensor afe

File: 13_sensor_afe.kicad_sch

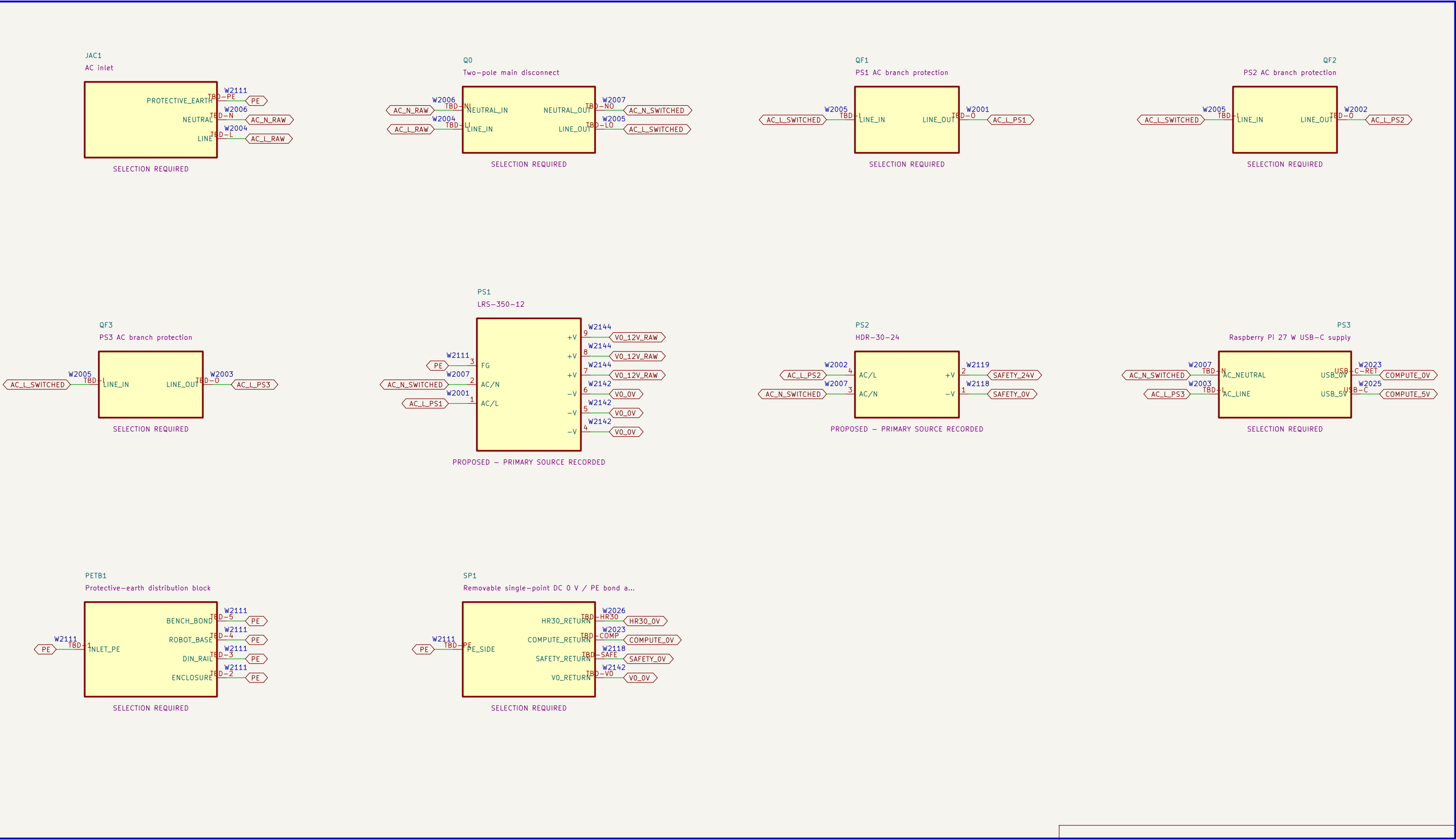
14 14 harness terminals

File: 14_harness_terminals.kicad_sch

V2.1 CONNECTED DESIGN INTENT – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR ENERGIZATION		
Project Button		
Sheet: /		
File: project-button-v2.kicad_sch		
Title: PB Electrical V2.1 – index		
Size: A3	Date: 2026-08-05	Rev: V2.1-P0.2
KiCad E.D.A. 10.0.5		Id: 1/15

01 AC input, protective earth, and separately protected DC domains

Mains branches feed the 12 V actuator, 24 V safety, and 5 V compute domains independently. PE is never switched. DC-to-PE links are open pending EMC and fault review.



NOTE 1: PE continuity is mandatory; SP1 is not populated by this preliminary design.

NOTE 2: PS1 output-terminal current sharing and conductor allocation require the terminal-current and thermal design.

NOTE 3: Before any permitted energization, inspect and control the LRS-350 115/230 V input-selector position. No fuse, breaker, or conductor value is released.

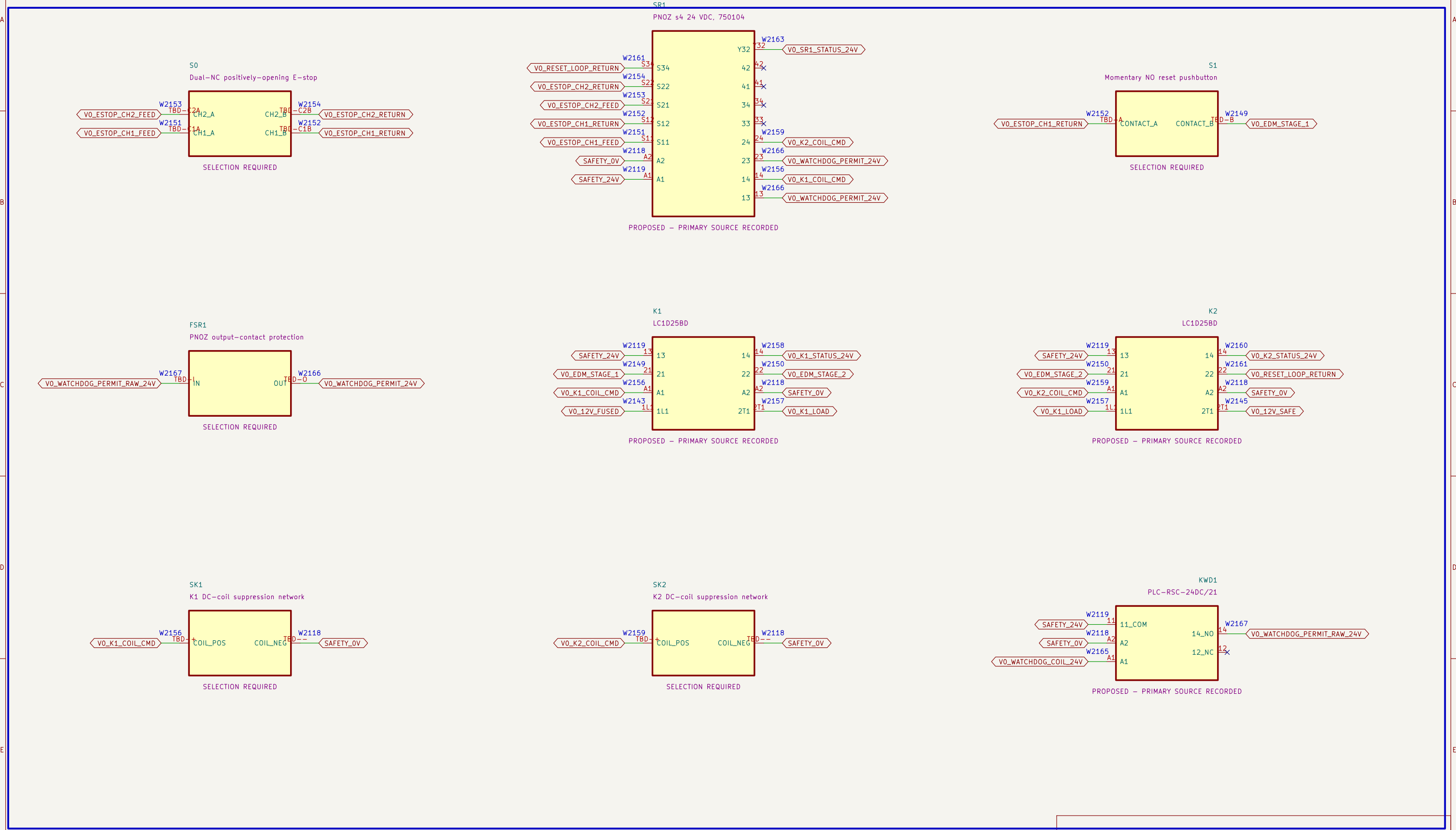
QUALIFIED REVIEW REQUIRED
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Project Button
Sheet: /01 01 ac pe power/
File: 01_ac_pe_power.kicad_sch

Title: PB Electrical V2.1 – 01 01 ac pe power

Size: A3	Date: 2026-08-05	Rev: V2.1-P0.2
KiCad E.D.A. 10.0.5		Id: 2/15

02 HR–VO dual–channel E–stop, falling–edge monitored start, watchdog permit, EDM, and redundant contactors

PNOZ s4 750104 is the proposed authoritative selection. Reset completes only the S12–reset–EDM–S34 start loop. It is not electrically connected to ARM or actuator commands.



NOTE 1: S12 – S1 – K1(21–22) – K2(21–22) – S34 is the proposed monitored start / EDM loop.

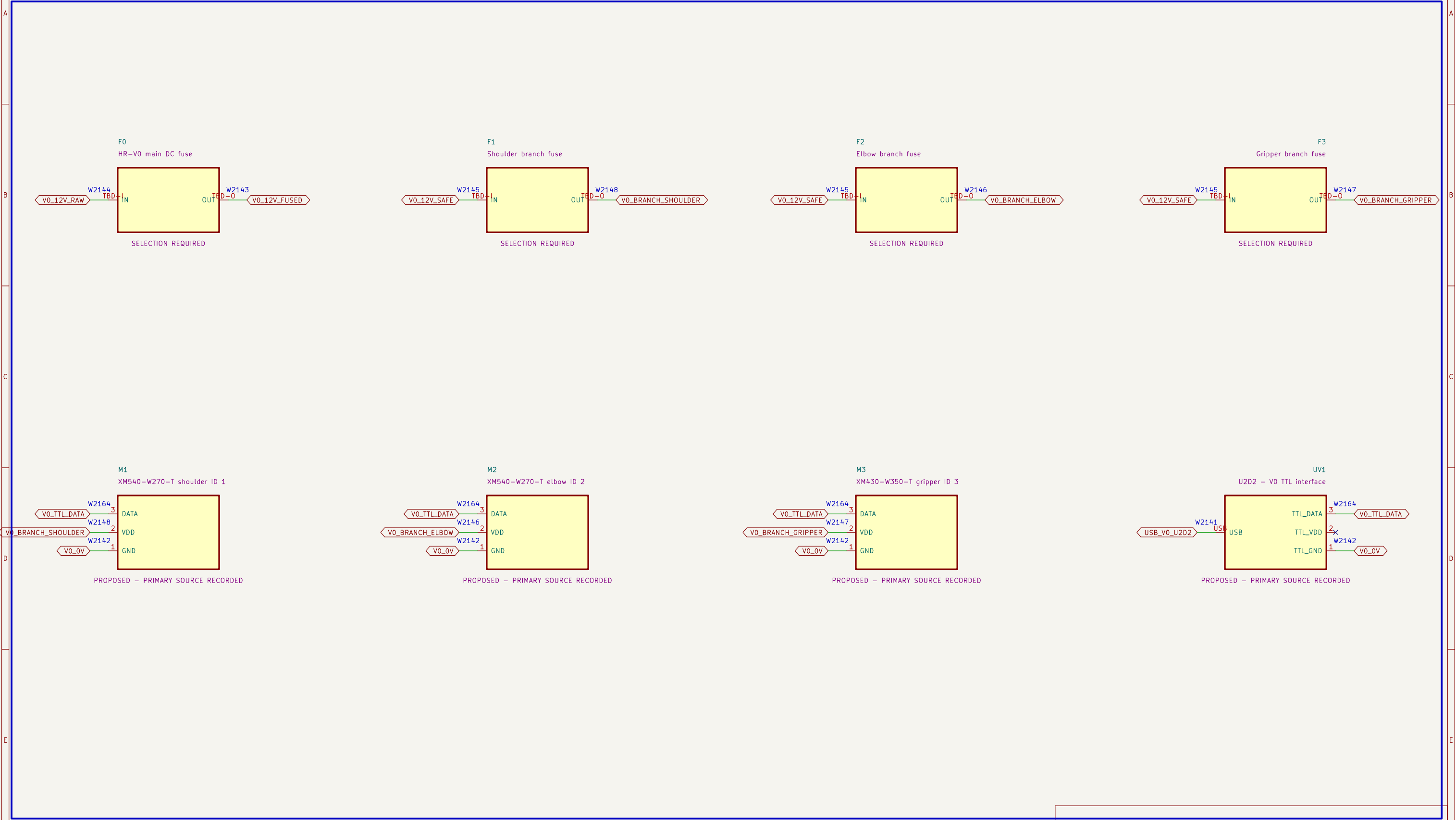
NOTE 2: Safety outputs 13–14 and 23–24 drive separate contactor coils; 33–34 is intentionally unused.

NOTE 3: The start/feedback loop has no relay diagnostic for shorts or cross–shorts; route it separately or protect it and justify fault exclusion. 41–42 and Y32 are diagnostic only.

NOTE 4: FSR1 and SK1/SK2 are mandatory selection gates. Verify output–contact protection, coil suppression, dropout, and stopping time before release.

03 HR–V0 actuator branch protection, star power injection, and one TTL data bus

Each actuator receives an individually protected power branch. The U2D2 participates in signaling only; actuator current is not summed through a U2D2 Power Hub.



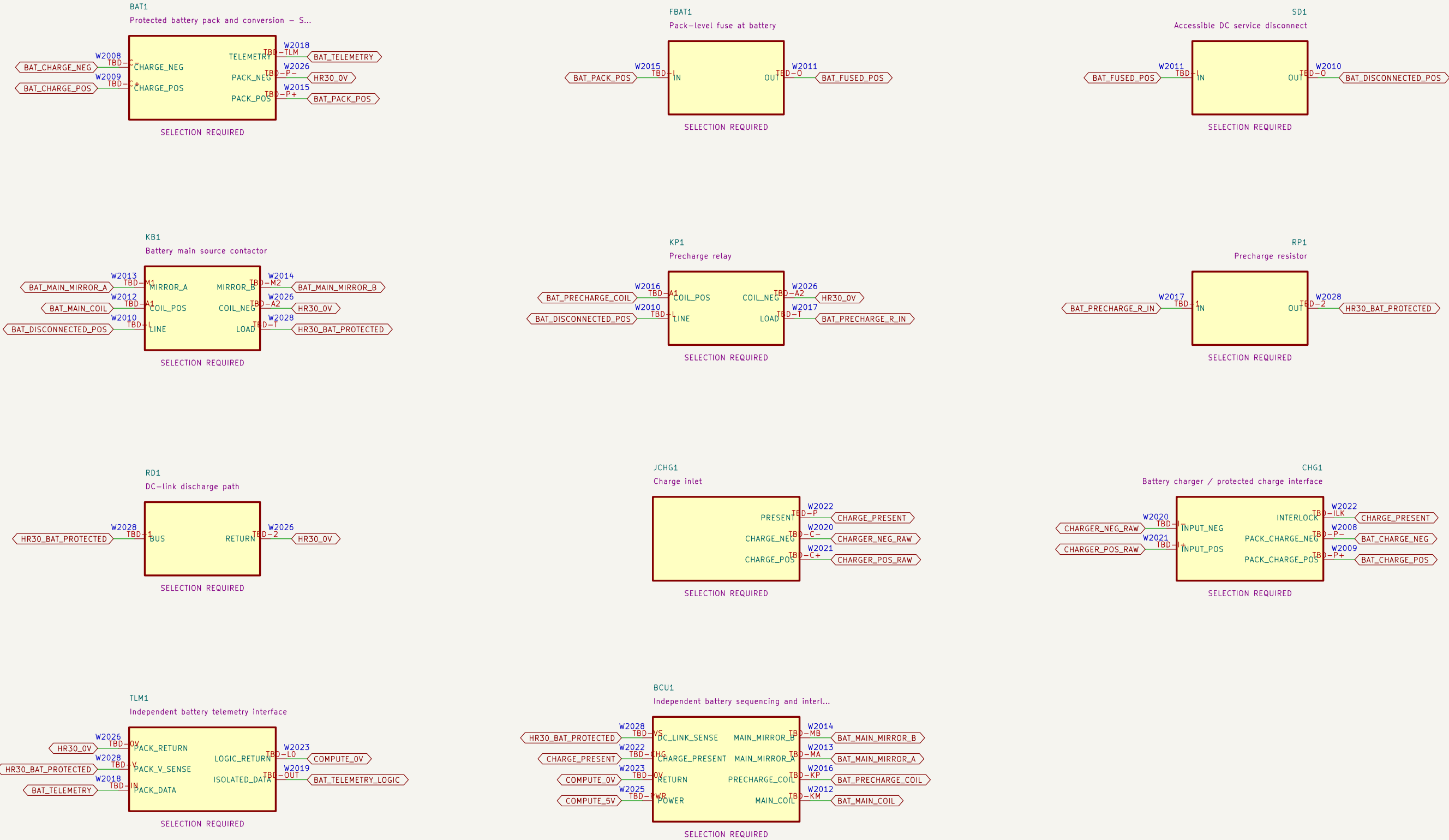
NOTE 1: Provisional IDs are configuration data, not connector pin numbers. Release requires a collision scan and recorded configuration hash.

NOTE 2: The former 20 A / 7.5 A / 7.5 A / 3 A values are not carried into V2 as approved selections.

NOTE 3: Standard fully populated 3–pin TTL cables carry VDD and would parallel the protected branches. Release a data–only harness/breakout and pass continuity plus no–backfeed tests.

05 HR–30W protected battery, service disconnect, parallel precharge, charger interlock, and telemetry

This is a functional topology only. Pack chemistry, voltage window, BMS ports, fuse, contactors, resistor, timing, thresholds, charger, and connectors remain unresolved.



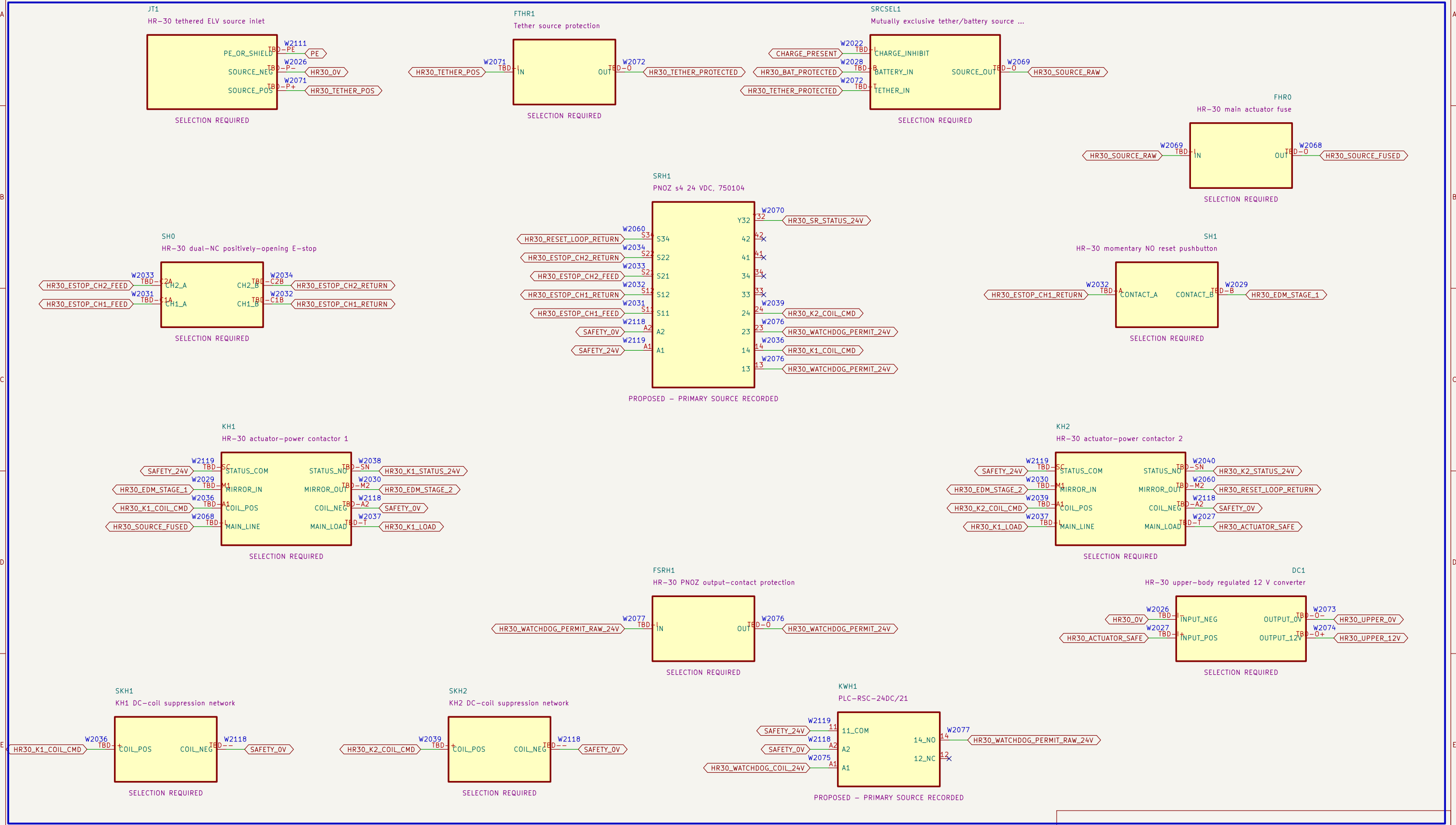
NOTE 1: KP1 plus RP1 is explicitly parallel to KB1. It is not a series element in the normal current path.

NOTE 2: Charge–present must inhibit actuator–power eligibility in hardware or verified safety control; implementation remains open.

QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR ENERGIZATION		
Project Button		
Sheet: /05 05 hr30 battery precharge/ File: 05_hr30_battery_precharge.kicad_sch		
Title: PB Electrical V2.1 – 05 05 hr30 battery precharge		
Size: A3	Date: 2026–08–05	Rev: V2.1–P0.2
KiCad E.D.A. 10.0.5		Id: 6/15

06 HR-30 source selection, dual-channel safety chain, redundant interruption, and upper-body conversion

Tether and battery are alternative protected sources and are not paralleled. HR-30 contactors are not assumed to be the HR-V0 LC1D25BD selection.



NOTE 1: HR-V0 and HR-30 are separate build variants. Their safety components are not installed or cross-wired as one assembly.

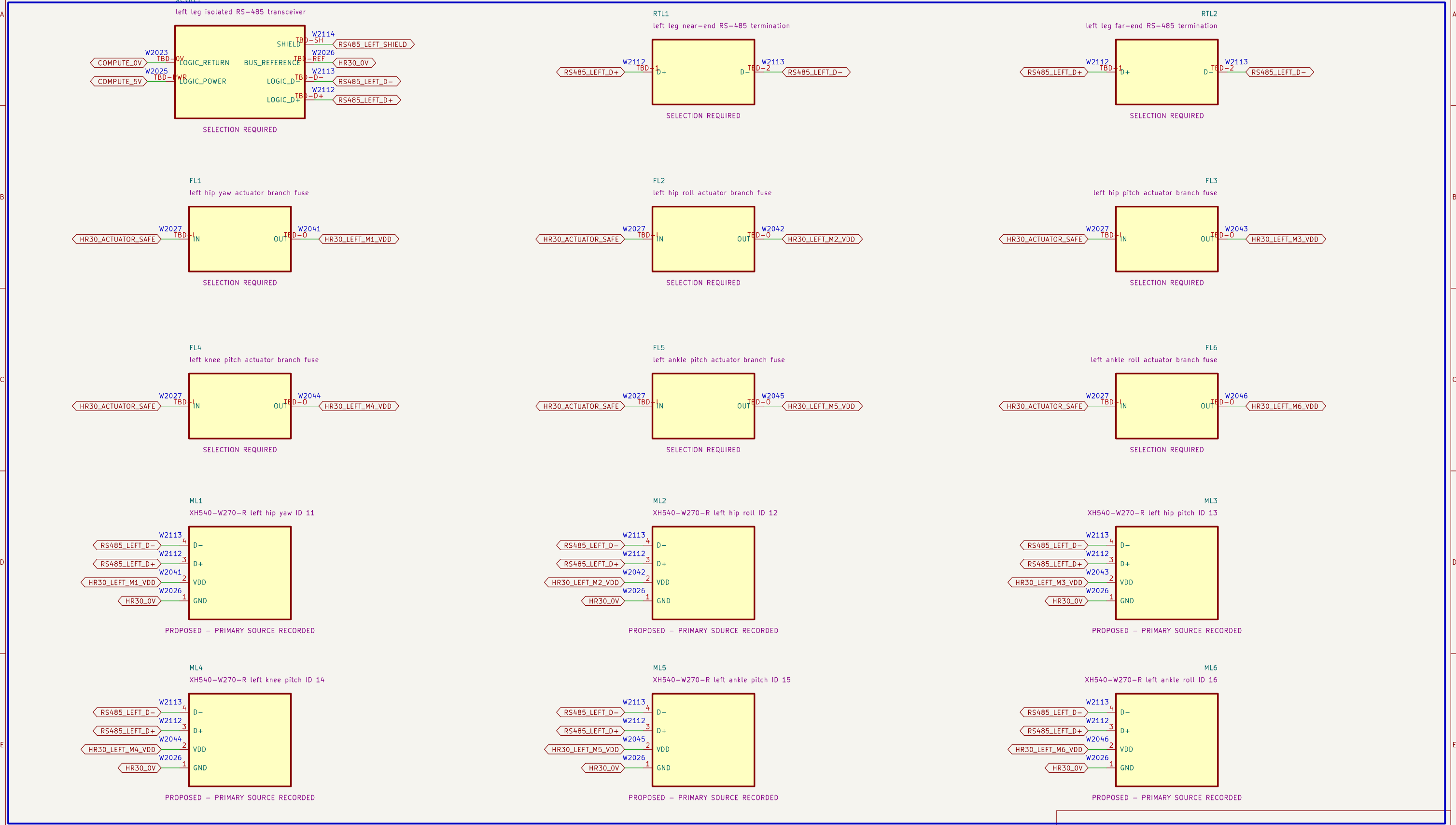
NOTE 2: SRH1 reset/EDM loop mirrors the V0 logic, but HR-30 contactor terminal numbers remain intentionally unresolved.

NOTE 3: The start/feedback loop has no relay diagnostic for shorts or cross-shorts; route it separately or protect it and justify fault exclusion. 41-42 and Y32 are diagnostic only.

NOTE 4: FSRH1 and SKH1/SKH2 are mandatory selection gates. Verify output-contact protection, coil suppression, dropout, and stopping time before release.

07 HR–30 left leg: six individually fused actuators and one linear RS–485 bus

Provisional IDs 11–16. Termination is present only at the controller and far physical ends; biasing belongs to the selected transceiver implementation.



NOTE 1: D+/D– must be routed as a controlled twisted pair. Shield/drain termination and chassis strategy are SELECTION REQUIRED.

NOTE 2: Power conductors are separately injected and routed away from the data pair; cable separation dimensions require harness review.

NOTE 3: Standard fully populated 4–pin RS–485 DYNAMIXEL cables carry VDD and would parallel protected branches. Release a VDD–isolating harness/breakout and pass continuity plus no–backfeed tests.

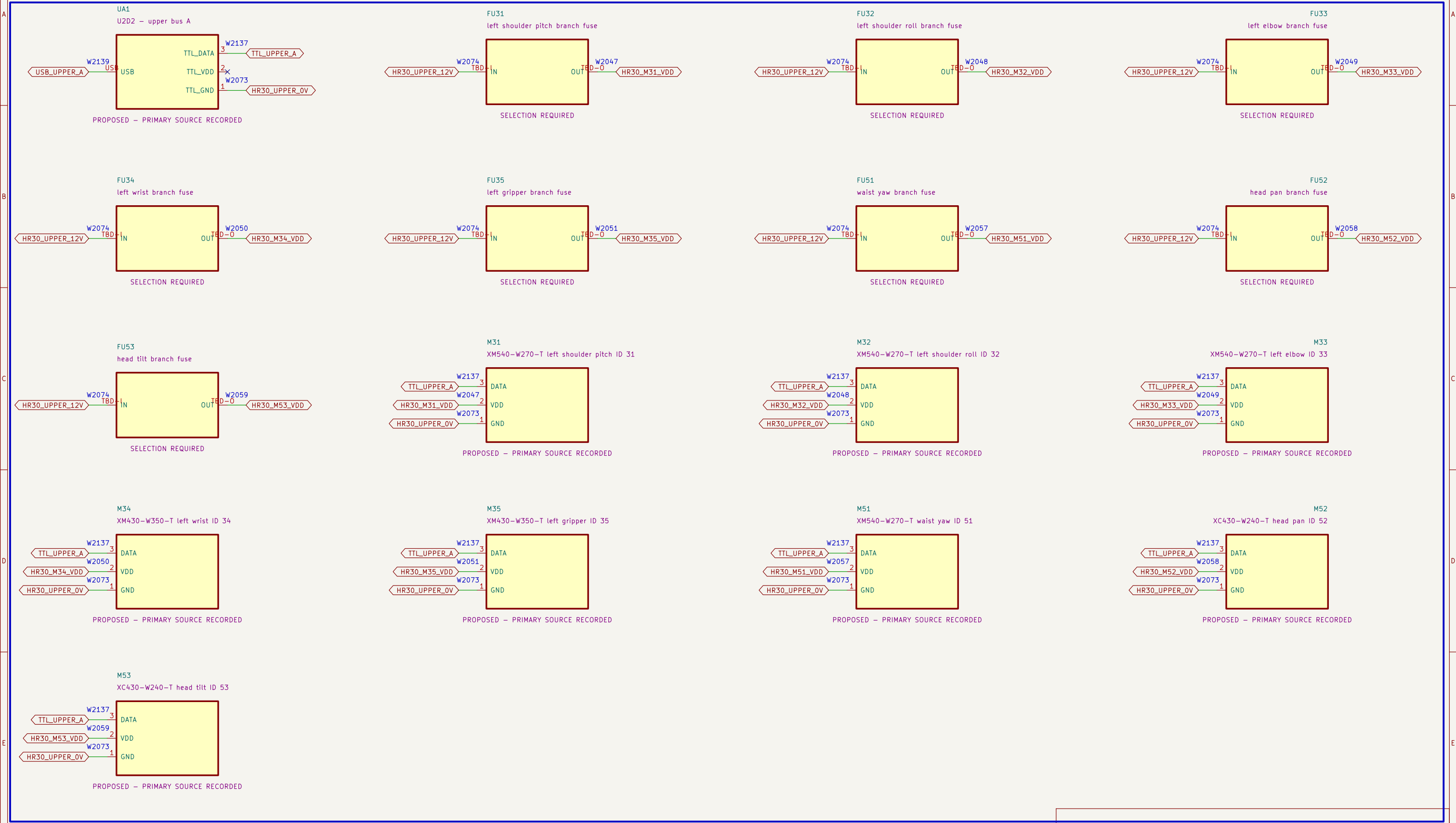
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Sheet: /07 07 left leg rs485/
File: 07_left_leg_rs485.kicad_sch

Title: PB Electrical V2.1 – 07 07 left leg rs485

Size: A3	Date: 2026–08–05	Rev: V2.1–P0.2
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09 HR-30 upper-body TTL bus A: left arm, waist, and head

One physical U2D2 serves one shared TTL chain. Eight actuator branches are individually protected and power-injected; no U2D2 Power Hub carries their summed current.



NOTE 1: This is one bus, not separate left-arm and head buses. A third bus would require a third physical interface and BOM revision.

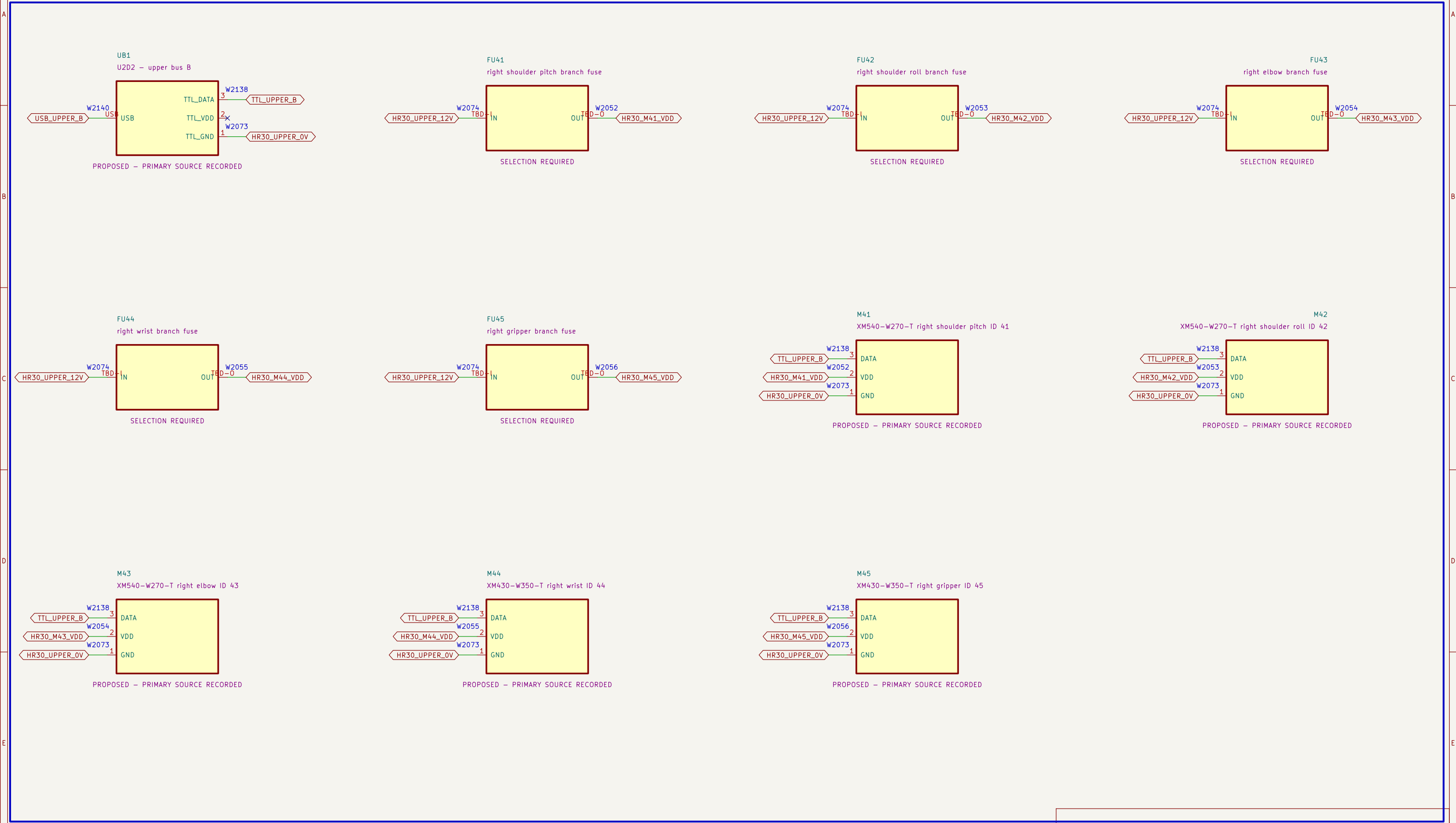
NOTE 2: Provisional IDs require collision scanning and configuration-hash capture before test.

NOTE 3: Standard fully populated 3-pin TTL cables carry VDD and would parallel protected branches. Release a data-only harness/breakout and pass continuity plus no-backfeed tests.

QUALIFIED REVIEW REQUIRED PRELIMINARY – NOT APPROVED FOR ENERGIZATION Project Button		
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Title: PB Electrical V2.1 – 09 09 upper bus a		
Size: A3	Date: 2026-08-05	Rev: V2.1–P0.2
KiCad E.D.A. 10.0.5		Id: 10/15

10 HR-30 upper-body TTL bus B: right arm

A second physical U2D2 serves the one right-arm TTL chain. Five actuator branches are individually protected and power-injected.



NOTE 1: One U2D2 is assigned to exactly one physical bus. No second bus is implied from UB1.

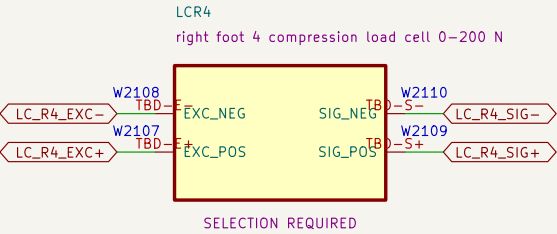
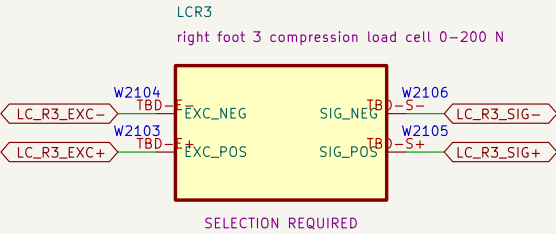
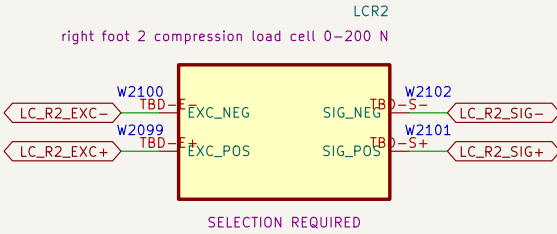
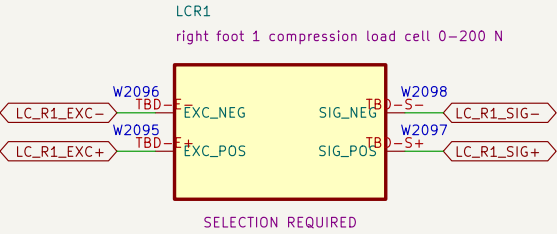
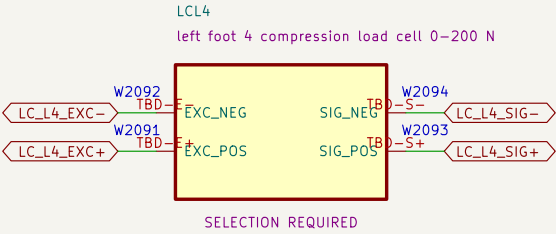
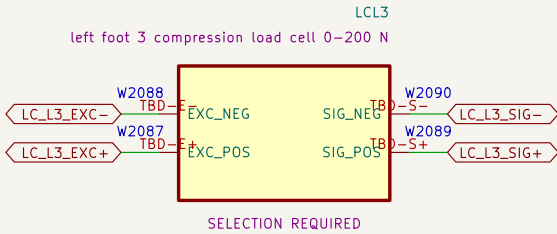
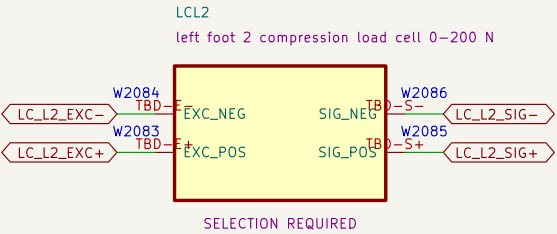
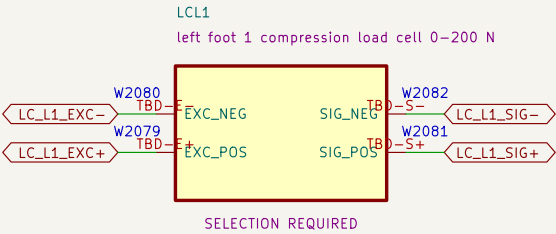
NOTE 2: Mating connectors, contacts, cable gauge, strain relief, and pin-orientation inspection remain open.

NOTE 3: Standard fully populated 3-pin TTL cables carry VDD and would parallel protected branches. Release a data-only harness/breakout and pass continuity plus no-backfeed tests.

QUALIFIED REVIEW REQUIRED PRELIMINARY – NOT APPROVED FOR ENERGIZATION Project Button		
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Size: A3	Date: 2026-08-05	Rev: V2.1–P0.2
KiCad E.D.A. 10.0.5		Id: 11/15

11 Foot load cells and unresolved bridge harness interfaces

Eight load cells are shown as functional bridge interfaces only. Exact parts, wiring colors, overload behavior, connectors, and calibration remain unresolved.



NOTE 1: Bridge excitation and signal pairs are logical nets; physical connector pins and cable colors are not assigned.

NOTE 2: Four load cells per foot remain the requirement baseline; overload and cross-axis tests are release gates.

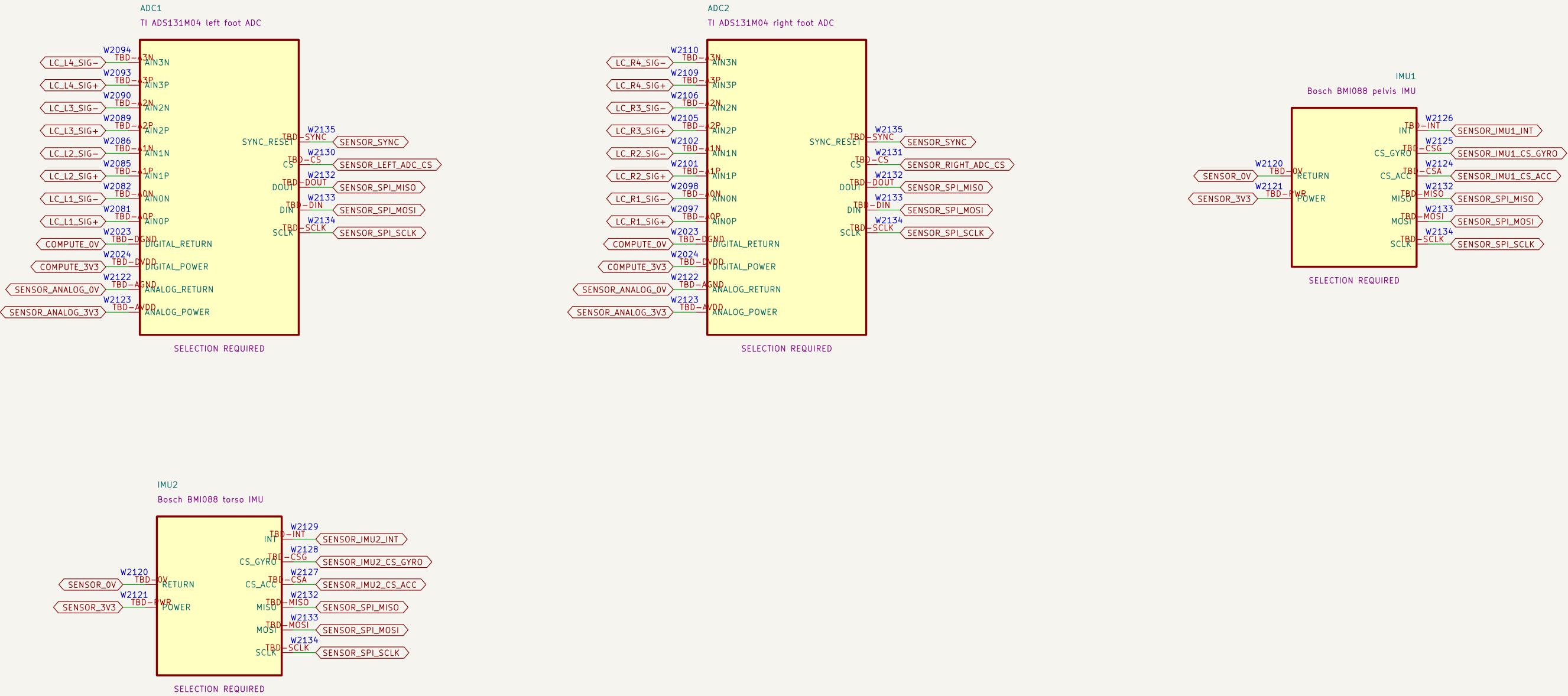
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File: 11_sensors_sync.kicad_sch

Title: PB Electrical V2.1 – 11 11 sensors sync

Size: A3	Date: 2026–08–05	Rev: V2.1–P0.2
KiCad E.D.A. 10.0.5		Id: 12/15

12 Foot ADCs, pelvis/torso IMUs, SPI interfaces, and synchronization

Proposed silicon is shown at functional–interface level only. No package pin map or manufacturable PCB circuit is asserted.



NOTE 1: ADS131M04 and BMI088 pin numbers are intentionally not asserted because board–level packages and integration are not selected.

NOTE 2: Sensor update–rate requirements remain verification targets, not evidence that this preliminary interface meets them.

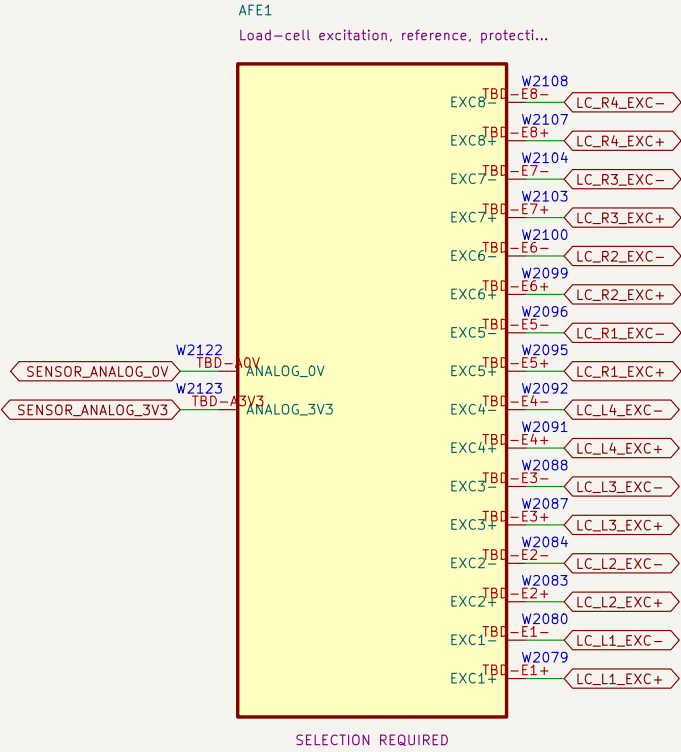
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Title: PB Electrical V2.1 – 12 12 sensor adc imu

Size: A3	Date: 2026–08–05	Rev: V2.1–P0.2
KiCad E.D.A. 10.0.5		Id: 13/15

13 Load–cell excitation, reference, protection, anti–alias filtering, and analog power

AFE1 is a functional boundary only. No circuit values, silicon, package pins, protection parts, grounding implementation, or PCB layout are selected.



NOTE 1: A complete low–noise analog design and error budget are required before any sensor PCB can be fabricated.

NOTE 2: Load–cell excitation, ADC reference, calibration, shielding, and analog return must be reviewed as one system.

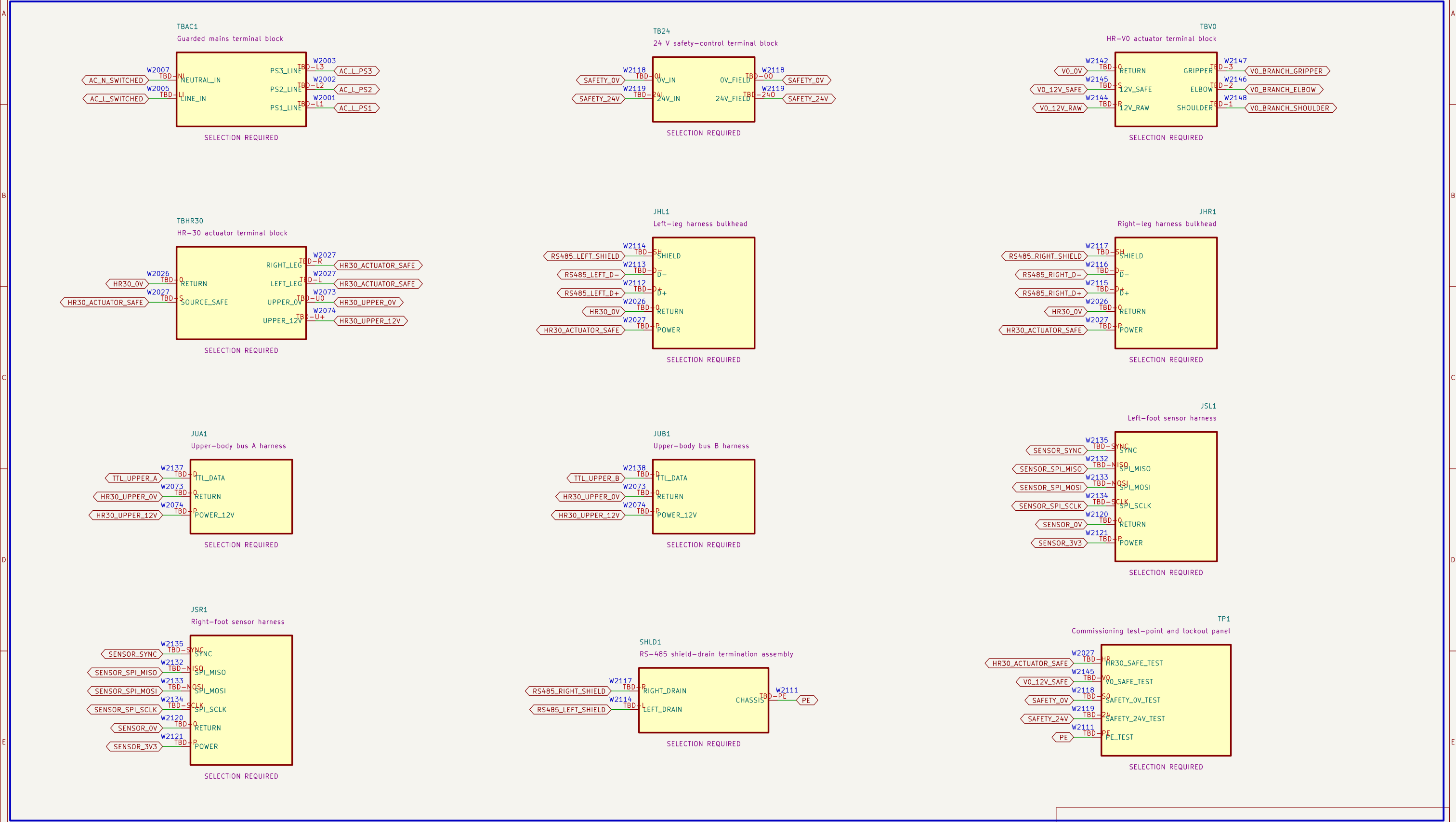
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Title: PB Electrical V2.1 – 13 13 sensor afe

Size: A3	Date: 2026–08–05	Rev: V2.1–P0.2
KiCad E.D.A. 10.0.5		Id: 14/15

14 Harness connectors, terminal blocks, shields, service separation, and unresolved physical pin maps

All pins on this sheet are logical interface names. Every terminal number, connector family, contact, wire gauge, color, and cable construction remains selection–dependent.



NOTE 1: No connector or terminal number on this sheet is physical. Every TBD identifier is an explicit release blocker.

NOTE 2: Minimum separation, creepage, clearance, shielding, bend radius, flex life, and service loops require the panel and harness design.